

The inner product $\langle x, y \rangle$ defines a metric. The group (G) acts on the set $\{X\}$. The absolute value $|x|$ is always non-negative. The interval $[a, b]$ contains all real numbers between a and b.

1. Prove the following directly from the two definitions and three axioms given in ICA2 on 8/20. You may not use any other results without proving them first. Every statement in your proof must be fully justified. See **Elements of (Proof) Style** guidelines in Canvas for information.
 - (a) Direct proof: For all integers n , if n is odd, then $9n + 5$ is even.
 - (b) Contrapositive proof: For all integers n , if $7n + 5$ is odd, then n is even.
 - (c) Proof by contradiction: For all integers n , if $3n + 2$ is even, then n is even.